

Employee Attrition Prediction Summary

This project analyzed employee information to identify factors associated with employees leaving the company. The analysis showed that attrition is influenced by several workplace factors rather than salary alone. Employees with lower job satisfaction, lower work-life balance, more overtime, fewer years at the company, and lower monthly income were more likely to leave.

Among different departments and job roles, Sales Representatives and Laboratory Technicians showed relatively higher employee turnover. Employees were also more likely to leave during their early years with the company, suggesting that retention efforts should focus on new employees.

The machine learning models were able to predict employee attrition with good accuracy. These predictions can help HR identify employees who may need additional support before they decide to leave.

Based on the findings, the company should improve employee engagement, strengthen work-life balance initiatives, conduct regular career development discussions, and provide targeted retention programs for high-risk employees. These actions can help reduce employee turnover and improve workforce stability.

The model should be used as a decision-support tool rather than replacing human judgment because personal reasons for leaving cannot always be captured by employee data.